2026

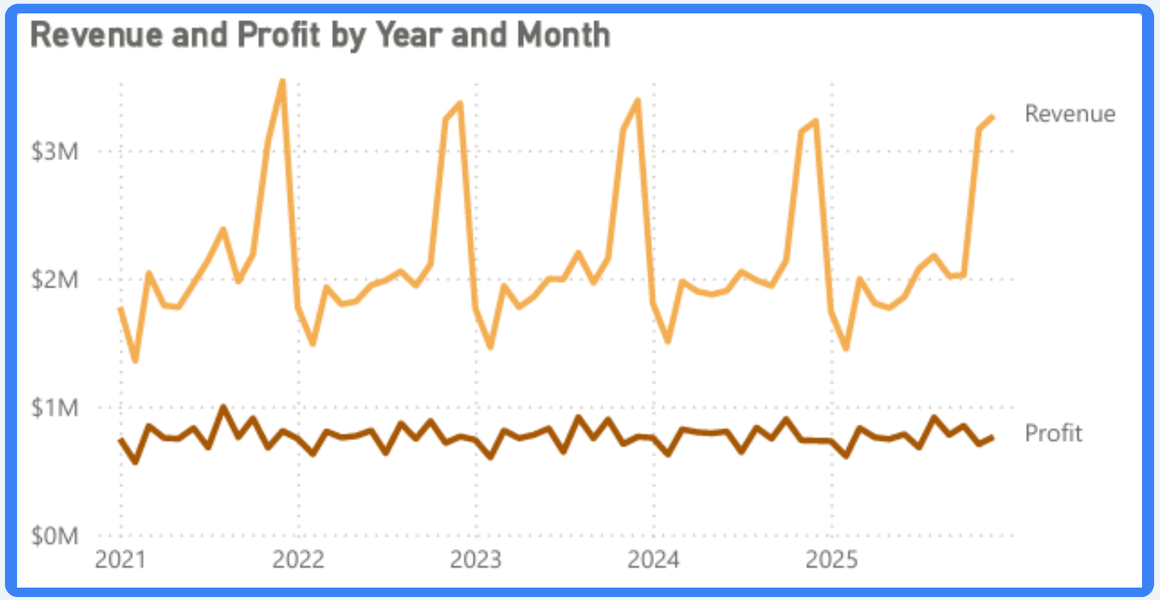
Discount Depth and Profit Margin: An Analysis of Year-End Sales Performance

E-Commerce Sales Analysis

By:
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Business Problem

The business sells online and runs heavy promotions every November-December. Revenue peaks in these two months every year from 2021 to 2025, but profit stays flat, so the extra sales are not turning into extra earnings. Management sees year-end revenue jump and expects profit to follow. It does not, monthly profit is lower in Nov-Dec than in Jan-Oct, even with more orders.

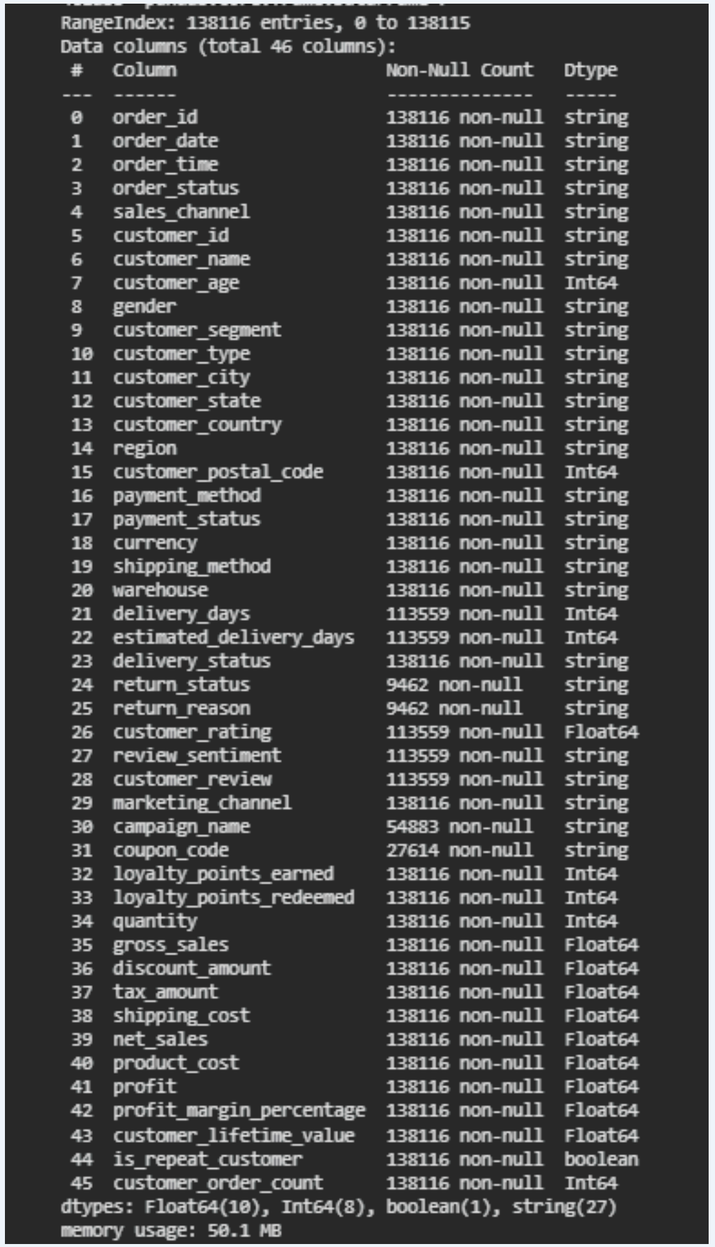


Business Objectives

1.
Explain why year-end revenue growth does not become profit growth

2.
Measure the gap: compare revenue, orders, profit, and margin in Nov-Dec against Jan-Oct

3.
Locate the causes, show how profit margin changes across discount bands



```
RangeIndex: 138116 entries, 0 to 138115
Data columns (total 46 columns):
 #   Column                    Non-Null Count   Dtype
---  ------                    --------------   -----
 0   order_id                  138116 non-null  string
 1   order_date                138116 non-null  string
 2   order_time                138116 non-null  string
 3   order_status              138116 non-null  string
 4   sales_channel             138116 non-null  string
 5   customer_id               138116 non-null  string
 6   customer_name             138116 non-null  string
 7   customer_age              138116 non-null  Int64
 8   gender                    138116 non-null  string
 9   customer_segment          138116 non-null  string
 10  customer_type             138116 non-null  string
 11  customer_city             138116 non-null  string
 12  customer_state            138116 non-null  string
 13  customer_country          138116 non-null  string
 14  region                    138116 non-null  string
 15  customer_postal_code      138116 non-null  Int64
 16  payment_method            138116 non-null  string
 17  payment_status            138116 non-null  string
 18  currency                  138116 non-null  string
 19  shipping_method           138116 non-null  string
 20  warehouse                 138116 non-null  string
 21  delivery_days             113559 non-null  Int64
 22  estimated_delivery_days   113559 non-null  Int64
 23  delivery_status           138116 non-null  string
 24  return_status             9462 non-null    string
 25  return_reason             9462 non-null    string
 26  customer_rating           113559 non-null  Float64
 27  review_sentiment          113559 non-null  string
 28  customer_review           113559 non-null  string
 29  marketing_channel         138116 non-null  string
 30  campaign_name             54883 non-null   string
 31  coupon_code               27614 non-null   string
 32  loyalty_points_earned     138116 non-null  Int64
 33  loyalty_points_redeemed   138116 non-null  Int64
 34  quantity                  138116 non-null  Int64
 35  gross_sales               138116 non-null  Float64
 36  discount_amount           138116 non-null  Float64
 37  tax_amount                138116 non-null  Float64
 38  shipping_cost             138116 non-null  Float64
 39  net_sales                 138116 non-null  Float64
 40  product_cost              138116 non-null  Float64
 41  profit                    138116 non-null  Float64
 42  profit_margin_percentage  138116 non-null  Float64
 43  customer_lifetime_value   138116 non-null  Float64
 44  is_repeat_customer        138116 non-null  boolean
 45  customer_order_count      138116 non-null  Int64
dtypes: Float64(10), Int64(8), boolean(1), string(27)
memory usage: 50.1 MB
```

Dataset

- **Total records:** 138,116 total transactions
- **Total features:** 46 columns
- **Periode:** 2021 - 2025
- **Data categories:**
  - Order
  - Customer profile
  - Geography
  - Payment
  - Shipping and logistics
  - Returns
  - Customer feedback
  - Marketing and promotion
  - Loyalty
  - Sales and financials
  - Customer value and behavior

Data Preparation



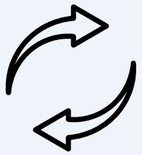
Duplicated data

No duplicate records were found: every **order_id** is unique across 138,116 orders, so no removal was needed.



Missing Value

Core sales, date and status fields are complete. Remaining gaps are structural and were kept: delivery and review fields are empty for orders that were cancelled, pending, and returned (24,557 rows), coupon and campaign fields are empty when not applicable.



Data validation and transformation

Formats were standardized (**order_date** to date; IDs and postal codes to text; amounts to decimals). **net_sales** was recalculated as **gross_sales** – **discount_amount**, and **profit** as **net_sales** – **product_cost**.



Feature Engineering

Added **order_sales** (**net_sales** + **tax_amount** + **shipping_cost**), **Disc %**, **Discount Band** (0–10% to 51–60%), **Year**, **Month No**, **Year-Month**, and **Period** (Jan–Oct vs Nov–Dec).



Data Filtering

Cancelled, Pending and **Returned** orders were excluded from the analysis, leaving only valid sales 113,559 orders .

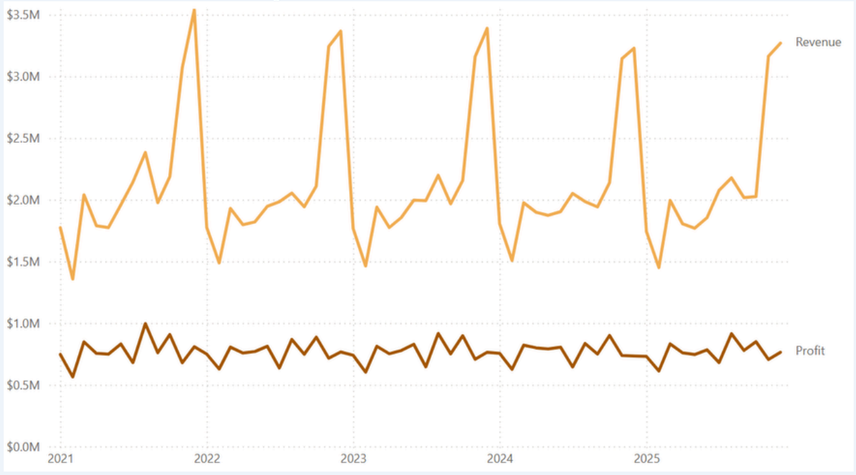


Feature Selection

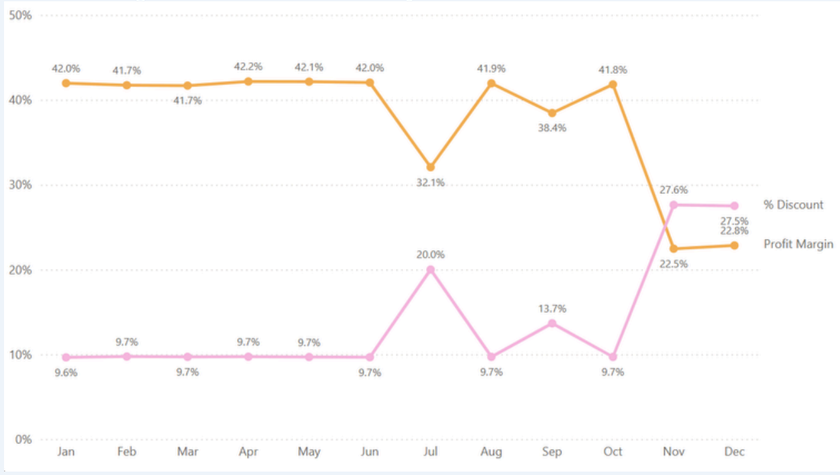
Kept fields tied to the discount–margin question: da**tes, status, sales** and **cost fields**, **discount**, and **period**. Personal data (**customer_name**, **customer_postal_code**) and **review text** were left out.

Margin Falls in the Same Months **Discounts Rise**

Revenue and Profit by Year and Month



Profit Margin and % Discount by Month

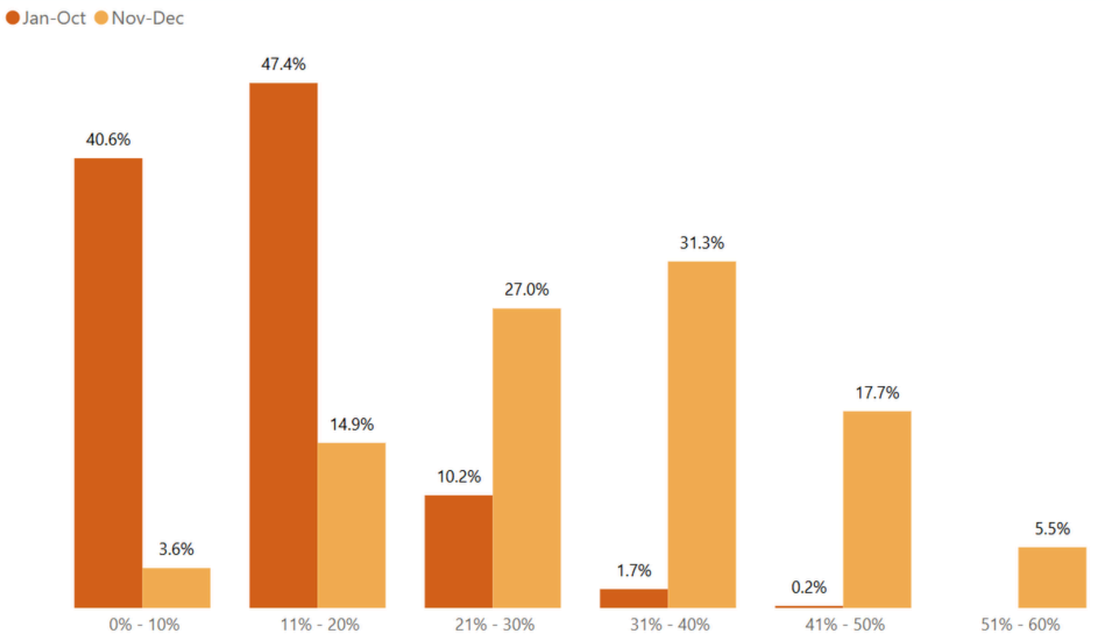


- Monthly revenue is up 70.8% in Nov–Dec, yet monthly profit is down 4.5%
- By calendar month, margin stays around 40% through the year and falls to about 22% in November and December, while the average discount rises from about 10% to about 27%
- July repeats the pattern: a 20% discount cuts margin to 32.1%.

Business Takeaway	Nov-Dec promotions lift revenue and orders, but profit does not follow, margin falls as the average discount rises from about 10% to about 27%

Nov - Dec Orders Shift to Deeper Discounts

Order Portion by discount band

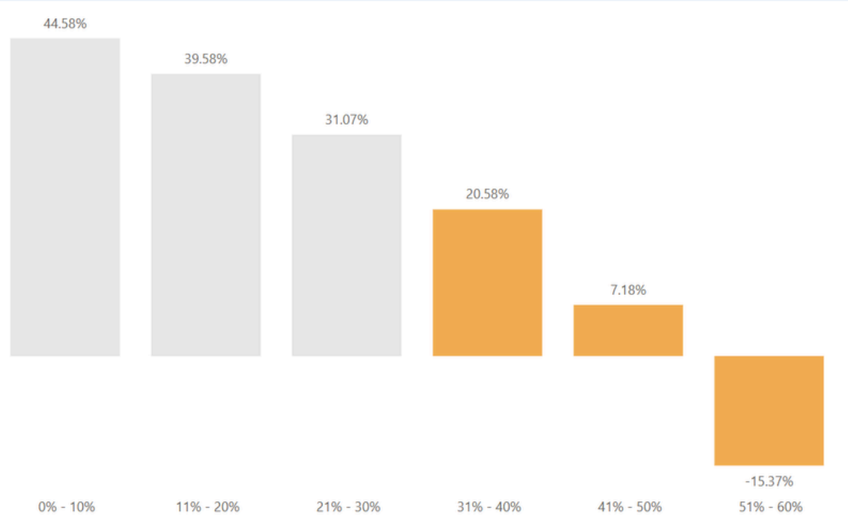


- Orders with 0-20% discount fall from 88% (Jan-Oct) to 18.5% (Nov-Dec).
- Orders above 30% discount rise from 1.9% (Jan-Oct) to 54.5% in (Nov-Dec).
- The 51-60% band, almost unused in Jan-Oct, reaches 5.5% of Nov-Dec orders.

Business Takeaway Nov-Dec promotions move most orders into discount bands that earn little margin

Margin Falls Sharply as Discount Deepens

Profit Margin by Discount Band



Discount Band	Total Revenue	Total Cost	Total Profit	Profit Margin
0% - 10%	$40,339,694	$22,355,223	$17,984,471	44.58%
11% - 20%	$47,946,731	$28,967,331	$18,979,400	39.58%
21% - 30%	$19,602,825	$13,511,734	$6,091,091	31.07%
31% - 40%	$13,329,925	$10,586,634	$2,743,292	20.58%
41% - 50%	$5,447,330	$5,056,184	$391,146	7.18%
51% - 60%	$1,240,501	$1,431,173	($190,672)	-15.37%

- Margin drops from 44.58% (0-10%) to 20.58% (31-40%) and turns negative (-15.37%) at 51-60%
- Bands above 30% hold 15.6% of revenue but only 6.4% of profit ($2.94M of $46.0M).
- Orders discounted above 30% keep only 14.7 cents of profit per dollar of revenue, against 39.9 cents for orders discounted up to 30%.

Business Takeaway Discounts above 30% give away most of the margin, and the 51–60% band loses money.

Conclusion

1. Year-end promotions raised monthly revenue 70.8% and orders 60.7%, but profit fell 4.5% and margin fell 17.86 points (40.51% → 22.65%)

2. The driver is discount depth: 54.5% of Nov-Dec orders carry a discount avoce 30% against 1.9% in Jan-Oct.

3. More orders added about $469K a month, but lower profit per order removed about $503K.



Recommendation

Priority	Recommendation	Expectation Business Impact
1	Stop discount above 50%	Remove the 51-60% band, which loses money (-$190,672 profit, -15.37% margin)
2	Cap the maximum discount at 30%	Avoid the 14.7% margin earned above 30% discount; protect year-end margin
3	Make 11-20% the default year-end depth	Keep margin near 39.58% while still offering a visible promotion
4	A/B test 15% vs 25% year-end discounts	Measure how many extra orders deeper discounts really win before the next campaign

Dashboard

SUMMARY

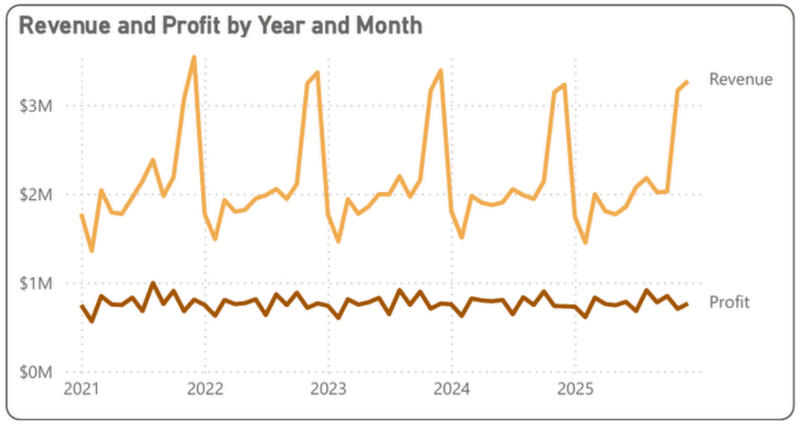
ANALYSIS

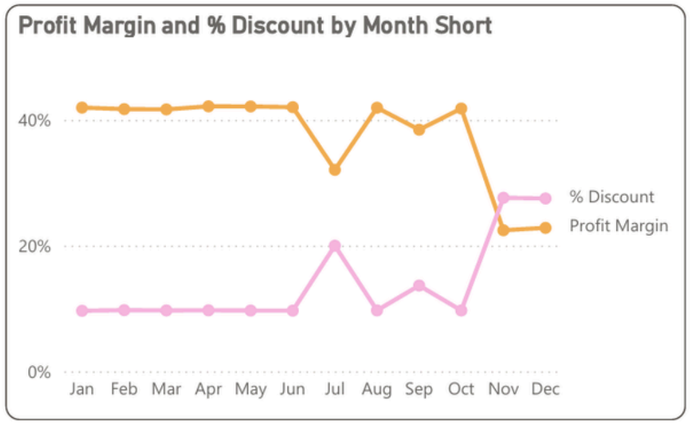
EXECUTIVE SUMMARY

Valid orders (only 'Completed' orders), 2021-2025. Monthly average Nov-Dec compared to Jan-Oct

Year: All

Revenue/Month Nov-Dec	Profit/Month Nov-Dec	Order/Month Nov-Dec	Margin/Month Nov-Des
$3.26M	$738K	2,763	22.65%
vs Jan-Oct $1.91M +70.82%	vs Jan-Oct $772K -4.50%	vs Jan-Oct 1,719 +60.74%	vs Jan-Oct 40.51% -17.86





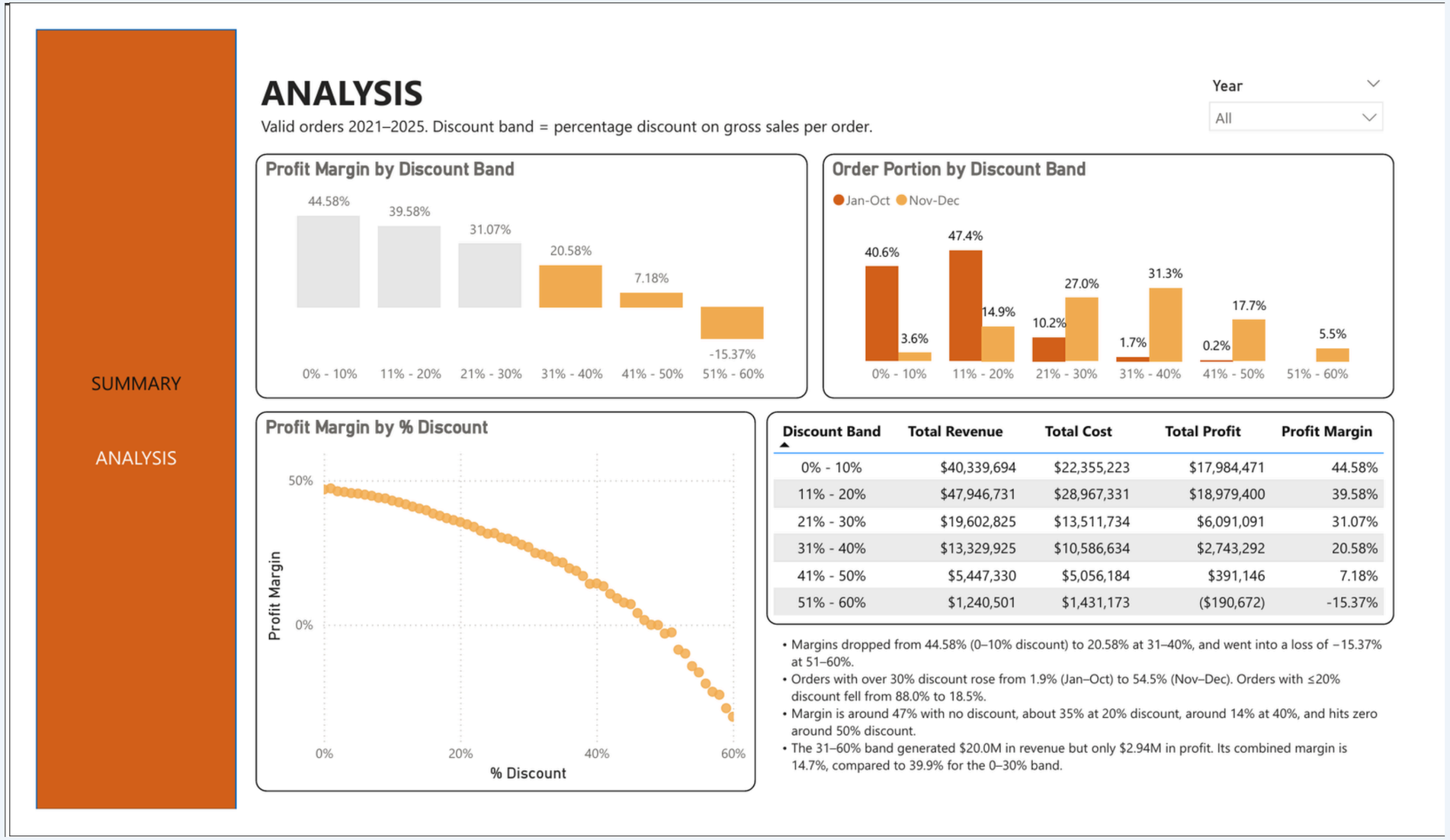
Findings: On average per month, revenue from Nov–Dec increase 70.8% ($1.91M → $3.26M) and orders went up 60.7% (1,719 → 2,763).

Impact: The extra revenue of about $1.35M per month didn't increase profit. Profit dropped 4.5% ($772K → $738K) and margin fell 17.86 points (40.51% → 22.65%).

Cause: Discounts went up from around 10% (Jan–Jun) to about 27% (Nov–Dec), so high volume was sold at thin margins.

ACCESS HERE

Dashboard



ACCESS HERE

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2026

Thank You



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